

EE2103-1 – ELECTROMAGNETIC FIELDS
(E&E)

Max. Marks: 100

❖ Use of ∇ operator table is allowed (refer last page of question paper).

20 Marks

- Two charges 1C and -4C exist in air. What is the direction of force?
A) Away from 1C
B) From 1C to -4C
C) Away from -4C
D) From -4C to 1C
- The relation between vector potential and field strength is given by
A) Gradient
B) Curl
C) Divergence
D) Del operator
- The Gaussian surface for a point charge will be
A) Sphere
B) Cube
C) Cuboid
D) Cylinder
- A hollow metallic sphere of radius r is kept at potential of 1 Volt. The total electric flux coming out of the concentric spherical surface of radius $R(>r)$ is
A) $4\pi\epsilon_0 r$
B) $4\pi\epsilon_0 r^2$
C) $4\pi\epsilon_0 R$
D) $4\pi\epsilon_0 R^2$
- Find the potential of the function $V = \frac{60\cos\theta}{r}$ at the point P(3, 60, 25).
A) 0.906
B) 18.126
C) 2.396
D) 10
- The electrostatic energy in an electric field does not depend on which of the following?
A) Magnitude of charges
B) Permittivity
C) Applied electric field
D) Flux lines
- The work done in moving a test charge from one point to another in an equipotential surface is
A) Zero
B) Infinity
C) Defined as the potential
D) 1J
- The capacitance of a material in air with area 20 units and distance between plates is 5cm.
A) 354.16pF
B) 35.416pF
C) 3.5416nF
D) 3.5416pF
- The magnetic field of a finite current element with 2A current and height $1/2\pi$ is
A) 1
B) 2
C) $1/2$
D) $1/4$
- Which of the following cannot be computed using the Biot Savart law?
A) Magnetic field intensity
B) Magnetic flux density
C) Electric field intensity
D) Permeability
- Find the inductance of a coil with turns 50, flux 3 units and a current of 0.5A
A) 8.33
B) 300
C) 150
D) 75
- The point form of Ampere law is given by
A) $\text{Curl}(\mathbf{B}) = \mathbf{J}$
B) $\text{Curl}(\mathbf{D}) = \mathbf{J}$
C) $\text{Curl}(\mathbf{V}) = \mathbf{J}$
D) $\text{Curl}(\mathbf{H}) = \mathbf{J}$

13. The value of $\oint \vec{H} \cdot d\vec{L}$ will be
 A) I B) J
 C) B D) H
14. Current is flowing in a conductor, the magnetic field strength at a radius r from the conductor is
 A) Radial to the conductor B) Parallel to the conductor
 C) Circumferential to the conductor D) Axial to the conductor
15. The unit for reluctance is
 A) Ohm B) Mho
 C) Wb/A D) A/Wb
16. A boundary of separation between two magnetic materials is identified by which factor?
 A) Change in the permeability B) Change in permittivity
 C) Change in magnetization D) Polarization
17. Find the ratio of permeability of the two media when the wave is incident on the boundary at 45 degree and reflected by the boundary at 60 degrees.
 A) 1:1 B) $\sqrt{3}:1$
 C) $1:\sqrt{3}$ D) $1:\sqrt{2}$
18. When a parallel plate capacitor is connected to AC source
 A) Conduction current is equal to the displacement current B) Conduction current is greater than the displacement current
 C) Conduction current is less than the displacement current D) Displacement current is zero
19. Find the characteristic impedance expression in terms of the inductance and capacitance parameters.
 A) $Z_o = \sqrt{LC}$ B) $Z_o = LC$
 C) $Z_o = \sqrt{L/C}$ D) $Z_o = \frac{1}{\sqrt{LC}}$
20. Which primary parameter is uniformly distributed along the length of the conductor?
 A) R B) L
 C) C D) G

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) State and explain Coulomb's law in complete vector form and there by obtain expression for electric field intensity $\vec{E}$ in Cartesian coordinate system.	08	L*2	1	2,3
	b) Given Electric field $\vec{E} = (\frac{1}{z^2})(8xyz\vec{a}_x + 4x^2z\vec{a}_y - 4x^2y\vec{a}_z)$ V/m. Find the differential amount of work done in moving a 60nC charge through a distance of 2 μ m, starting at P(2,-2,3) and proceeding in the direction. (i) $\frac{6}{7}\vec{a}_x - \frac{3}{7}\vec{a}_y - \frac{2}{7}\vec{a}_z$ (ii) $\frac{3}{7}\vec{a}_x + \frac{6}{7}\vec{a}_y$	08	L3	2	2,3
2.	a) Find $\vec{E}$ at a point P(1,1,1) caused by 4 identical 3nC charges located at P1(1,1,0), P2(-1,1,0), P3(-1,-1,0) and P4 (1,-1,0).	06	L3	1	1,2
	b) Find the approximate value for the total charge enclosed in an incremental volume of $10^{-9}m^3$ located at the origin, if $\vec{D} = e^{-x}\sin y\vec{a}_x - e^{-x}\cos y\vec{a}_y + 2z\vec{a}_z$ C/m ² .	04	L3	1	2,3
	c) Define Potential and Potential Difference. Explain the potential field of a point charge. Justify the field is a scalar field.	06	L1 L4	2	2,3
3.	a) Explain the application of Gauss's Law to differential volume element and hence define the divergence.	06	L2	1	1,2
	b) Define and explain current and current density. Derive the point form of continuity equation.	06	L1	2	1,2
	c) Find div $\vec{D}$ at the origin if $\vec{D} = e^{-x}\sin y\vec{a}_x - e^{-x}\cos y\vec{a}_y + 2z\vec{a}_z$.	04	L3	2	2,3

Unit – II

4. a) From fundamentals derive the Laplace's equation. Using Laplace's equation, find an expression for capacitance of a parallel plate capacitance. 06 L2 3 1,2
 b) State and explain Stoke's theorem. 05 L1 3 1,2
 c) Consider a parallel plate capacitor having $d=5\text{mm}$, $S = 80\text{cm}^2$ and $\epsilon_r = 10$.
 (i) find the capacitance,
 (ii) If a DC voltage of 50V is applied across the capacitor, calculate $\vec{E}$, D , Q and total stored energy. 05 L3 3 2,3
5. a) A current filament carrying 15A in the $\vec{a}_z$ direction lies along the entire z- axis. Find $\vec{H}$ in rectangular coordinates at:
 (i) $P_A(\sqrt{20}, 0, 4)$ (ii) $P_B(2, -4, 4)$. 06 L3 3 2,3
 b) Derive an expression for inductance per meter length of a coaxial of inner radius 'a' and outer radius 'b' from the basics. 06 L2 4 1,2
 c) Obtain the magnetic boundary conditions at the interface between two magnetic materials. 04 L2 4 1,2
6. a) Define Ampere's Circuital Law. How can it help in determining magnetic field intensity due to a filamentary conductor carrying a current? 08 L2 3 1,2
 b) The field $\vec{B} = -2\vec{a}_x + 3\vec{a}_y + 4\vec{a}_z \text{ mT}$ is present in free space. Find the vector force exerted on a straight wire carrying 12A in the $\vec{a}_{AB}$ direction, given $A(1,1,1)$ and: i) $B(2,1,1)$ ii) $B(3,5,6)$. 08 L3 4 2,3

Unit – III

7. a) Show that the displacement current equals conduction current in the case of a parallel plate capacitor connected across an alternating source. 08 L2 4 1,2
 b) Derive the general transmission line equation. Obtain an expression for wave propagation in lossless medium. 08 L1 5 1,2
8. a) Given $\mu = 10^{-15}\text{H/m}$, $\epsilon = 4 \times 10^{-9}\text{F/m}$, $\sigma = 0$, and $\rho_v = 0$. Find k so that the following pair of fields satisfies Maxwell's equations:
 $D = 6\vec{a}_x - 2y\vec{a}_y + 2z\vec{a}_z \text{ nC/m}^2$ and
 $H = kx\vec{a}_x + 10y\vec{a}_y - 25z\vec{a}_z \text{ A/m}^2$. 05 L3 4 2,3
 b) Explain the concept of lossless propagation of sinusoidal voltage. 05 L2 5 1,2
 c) Explain maxwell's equation in point from (any three). 06 L1 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

∇ Operator Table

Divergence

Rectangular	$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$
Cylindrical	$\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$
Spherical	$\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(D_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$

Gradient

Rectangular	$\nabla V = \frac{\partial V}{\partial x} \vec{a}_x + \frac{\partial V}{\partial y} \vec{a}_y + \frac{\partial V}{\partial z} \vec{a}_z$
Cylindrical	$\nabla V = \frac{\partial V}{\partial \rho} \vec{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \vec{a}_\phi + \frac{\partial V}{\partial z} \vec{a}_z$
Spherical	$\nabla V = \frac{\partial V}{\partial r} \vec{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \vec{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \vec{a}_\phi$

Laplacian

Rectangular	$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$
Cylindrical	$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$
Spherical	$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right)$

Curl

Rectangular	$\nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \vec{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \vec{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \vec{a}_z$
Cylindrical	$\nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \vec{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \vec{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right] \vec{a}_z$
Spherical	$\nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \vec{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial(r H_\phi)}{\partial r} \right) \vec{a}_\theta$ $+ \frac{1}{r} \left[\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right] \vec{a}_\phi$
